

# GEOMETRIC PHASE, TORSION, AND THE MAGNETIC CONFINEMENT OF CHARGED PARTICLES

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**ABSTRACT.** The torsion of a curve in Euclidean space quantifies how much the curve twists. Geometric phase is a phenomenon that can explain how a cat falling with no angular momentum can still land on its feet. This paper investigates the relationship between the torsion of a curve and the concept of geometric phase to bring insight into how a magnetic field can be shaped to confine a charged particle in a fixed domain of Euclidean space.

## BACKGROUND

### What is ‘Geometric Phase’?

- *Geometric Phase*, i.e. holonomy, is the amount of information lost/alterd when a system undergoes some evolution that brings it back to its original state, after parallel transporting around a cycle on a manifold in some parameter space.
- *Dynamical phase*, on the other hand, describes how much information is lost/alterd after evolving a system with *time*.
- *What is necessary to define a geometric phase term?*
  - (1) A parameter space
  - (2) A fibre bundle over the parameter space,
  - (3) A connection (on the bundle)
  - (4) A defined closed curve in the parameter space

**Generalisation of Derivatives.** <sup>1</sup> The motivation here is that a familiar multivariable calculus concept of ‘directional derivative’ does not generally exist on arbitrary (smooth) manifolds. So a generalisation is required.

**Definition 0.1** (Directional Derivative). A **directional derivative** of a vector fields is a map  $\nabla : \text{Vec}(M) \times \text{Vec}(M) \rightarrow \text{Vec}(M)$  as  $(X, Y) \rightarrow \nabla_X Y$  such that  $\nabla$  is bilinear over  $\mathbb{R}$  and satisfies the product rule,

$$\nabla_X(fY) = D_X f \cdot Y + f \cdot \nabla_X Y$$

for all  $f \in C^\infty(M)$  and  $D$  is the well-defined directional derivative of  $f$ .

If  $\nabla$  satisfies  $\nabla_{fX} Y = f \cdot \nabla_X Y$  then  $\nabla$  is *covariant* or *tensorial w.r.t. direction* (i.e. linear w.r.t. scalar multiplication by a smooth function).

But there is no natural choice of covariant derivative of vector fields for arbitrary manifolds. The workaround is the Lie derivative.

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<sup>1</sup>[This article](#) was helpful for this section

**Definition 0.2** (Lie Derivatives). The **Lie derivative** of a vector field  $Y \in \text{Vec}(M)$  in the direction of  $X$  is the vector field defined by,

$$(\mathcal{L}_X Y)(p) := \lim_{t \rightarrow 0} \frac{Y_p - (\rho_{t*} Y)_p}{t}$$

for all  $p \in M$ , where  $\rho : M \rightarrow M$  is the **flow** of  $X \in \text{Vec}(M)$  defined as

$$\left. \frac{d}{dt} \rho_t(p) \right|_{t=0} = X_p.$$

$\rho_{t*} Y$  is the *pushforward* of the vector field  $Y$  by the diffeomorphism  $\rho_t$  defined by  $(d\rho_t)_{\rho_t^{-1}(p)} Y_{\rho_t^{-1}(p)}$ .

For example, on an  $\mathbb{R}^n$  manifold, the natural choice for a covariant directional derivative can be  $\nabla : \text{Vec}(\mathbb{R}^n) \times \text{Vec}(\mathbb{R}^n) \rightarrow \text{Vec}(\mathbb{R}^n)$  by

$$\nabla_X Y(p) := \lim_{t \rightarrow 0} \frac{Y_{p+tX_p} - Y_p}{t}$$

The definition of this natural covariant directional derivative for  $\mathbb{R}^n$  highlights the problem with defining one on an arbitrary manifold. For an arbitrary manifold, tangent vectors at different points can't be naturally compared. There is no natural connection between different tangent spaces. This is what a **connection** will provide.

### Fibre Bundles & connections. :

**Definition 0.3** (Fibre Bundle). Suppose  $M$  (“base/parameter space”),  $F$  (“fibre”), and  $E$  (“total space”) are all topological spaces. A **fibre bundle** over  $M$  with fibre  $F$  is the tuple  $(M, F, E, \pi)$  where  $\pi : E \rightarrow M$  is a continuous map such that  $E$  is locally the product of  $M$  and  $F$ .

$E$  is a collection of copies of  $F$  parametrised by the points of  $M$ .

Let  $\pi : E \rightarrow M$  be a smooth fibre bundle projection with fibre  $F$ . For each  $u \in E$ , let  $V_u$  denote the subspace of  $\mathbb{T}_u E$  consisting of vectors tangent to the fibre  $F_{\pi(u)}$ . Then  $V_u = \mathbb{T}_u(E_{\pi(u)}) = \ker(d\pi_u : \mathbb{T}_u E \rightarrow \mathbb{T}_{\pi(u)} M)$ .  $V_u$  is called the **vertical subspace** of  $u$ .

**Definition 0.4** (Ehresmann Connection). An **Ehresmann connection** on  $\pi : E \rightarrow M$  is a collection  $\Gamma = \{H_u | u \in E\}$  of vector subspaces  $H_u \subset \mathbb{T}_u E$  called the **horizontal subspace** such that

- (1)  $u \rightarrow H_u$  depends smoothly on  $u \in E$ ,
- (2)  $\forall u \in E$ , we have  $\mathbb{T}_u E = H_u \oplus V_u$ .

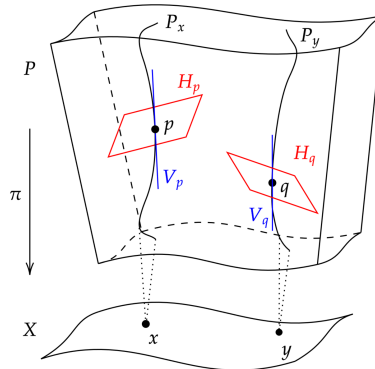


FIGURE 1. Cartoon of a fiber bundle quadruple  $(P, X, \pi, F)$

**Definition 0.5** (Principal  $G$ -Bundle). A **principal  $G$ -Bundle** over a manifold  $M$  ( $G$  a Lie group) is a fibre bundle  $\pi : P \rightarrow M$  with fibre  $G$  together with a smooth right action  $R$  of  $G$  on  $P$  such that:

- $R_g$  preserves fibres, and acts transitively + freely. i.e. on a fibre  $P_p$ , for points  $u_1, u_2 \in P_p$ ,  $\exists! g \in G$  such that  $R_g(u_1) = u_2$ .
- There is an open cover  $U_\alpha$  of  $M$  with local trivialisations  $\{\phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times G\}$  such that the action of  $G$  on  $\pi^{-1}(U_\alpha)$  under  $\phi_\alpha$  look like right multiplication. i.e. if  $\phi_\alpha(u) = (\pi(u), a)$ , then  $\phi_\alpha(R_g(u)) = (\pi(u), a \cdot g)$ .

Here a **right action**  $R$  of  $G$  on a manifold  $P$  is a group *anti*-homomorphism  $G \rightarrow \text{Diff}(P)$ ,  $g \rightarrow R_g$  such that  $P \times G \rightarrow P$  given by  $(u, g) \rightarrow R_g(u)$  is smooth.

**Definition 0.6** (Principal Bundle Connection).  $\pi : P \rightarrow M$  is a  $G$ -principal bundle. A **principal bundle connection**  $\Gamma$  on this bundle is a fibre bundle connection  $\Gamma = \{H_u \mid u \in P\}$  such that the  $G$ -action  $R$  preserves horizontal subspaces. i.e.

$$H_{R_g(u)} = (dR_g)_u(H_u)$$

for all  $g \in G$ .

**Theorem 0.7.**  $\pi : P \rightarrow M$  a principal  $G$ -fibre bundle, and  $\Gamma$  a principal fibre bundle connection. If  $p \in M$  and  $c : [0, 1] \rightarrow M$  is a smooth path in  $M$  with  $c(0) = p$ , then for each choice of lift  $u \in P_p$ , there exists a unique horizontal lift  $\tilde{c} : [0, 1] \rightarrow P$  of  $c$  with  $\tilde{c}(0) = u$ .

By moving through all  $u \in P_p$ , a map  $\tau_t : P_p \rightarrow P_{c(t)}$  between fibres, called the **parallel displacement/transport** along the curve  $c$  is obtained.

Equivalently, a fibre bundle connection can be defined by a vector bundle homomorphism.

**Definition 0.8** (Vector Bundle Homomorphism). A **vector bundle homomorphism**  $v : \mathbb{T}E \rightarrow \mathbb{T}E$  such that  $v^2 = v$  and  $v(\mathbb{T}_u E) = V_u$  for each  $u \in E$ . The corresponding horizontal subspace is  $H_u := \ker v \mid_{\mathbb{T}_u E}$

**Definition 0.9** (Vector Bundle). A **rank  $k$  vector bundle** over manifold  $M$  is a fibre bundle with the projection map  $\pi_E : E \rightarrow M$  with fibre  $\mathbb{R}^k$  such that each fibre  $E_p := \pi_E^{-1}(p)$  has the structure of a vector space with the following local linearity condition.

For all  $\omega \in E$  there is an open neighbourhood  $U$  of  $\pi_E(\omega)$  in  $M$  and a trivialisation map  $\phi : \pi_E^{-1}(U) \rightarrow U \times \mathbb{R}^k$  such that the map  $E_{\pi_E(\omega)} \rightarrow \mathbb{R}^k$  given by the composition

$$\omega' \rightarrow \phi(\omega') = (\pi_E(\omega), \xi) \rightarrow \xi$$

is a *linear isomorphism*.

And so now we can provide a definition of an *affine connection* over a tangent bundle.

**Definition 0.10** (Affine Connections). An **affine connection** on smooth manifold  $M$  with tangent bundle  $TM$  is a bilinear map  $\nabla : \text{Vec}(TM) \times \text{Vec}(TM) \rightarrow \text{Vec}(TM)$  that satisfies the following conditions for all  $X, Y \in \text{Vec}(TM)$ ,  $\forall f \in C^\infty(M)$  :

- Linearity:  $\nabla_{fX} Y = f \nabla_X Y$
- Leibniz's Rule:  $\nabla_X (fY) = D_X f \cdot Y + f \cdot \nabla_X Y$

### The Choice of a Frame. ,

**Definition 0.11** (Frame Field). Vector fields  $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$  on  $\mathbb{R}^3$  constitute a **frame field** on  $\mathbb{R}^3$  if

$$\mathbf{E}_i \cdot \mathbf{E}_j = \delta_{ij} \quad (1 \leq i, j \leq 3)$$

A suitable appropriate frame field must express the covariant derivatives of vector fields in terms of the vector fields themselves.

**Definition 0.12** (Connection Forms). Suppose  $\mathbf{v}$  is an arbitrary tangent vector at point  $\mathbf{p}$ . Then

$$\begin{pmatrix} \nabla_{\mathbf{v}} E_1 \\ \nabla_{\mathbf{v}} E_2 \\ \nabla_{\mathbf{v}} E_3 \end{pmatrix} = \begin{pmatrix} \omega_{11}(\mathbf{v}) & \omega_{12}(\mathbf{v}) & \omega_{13}(\mathbf{v}) \\ \omega_{21}(\mathbf{v}) & \omega_{22}(\mathbf{v}) & \omega_{23}(\mathbf{v}) \\ \omega_{31}(\mathbf{v}) & \omega_{32}(\mathbf{v}) & \omega_{33}(\mathbf{v}) \end{pmatrix} \begin{pmatrix} E_1(\mathbf{p}) \\ E_2(\mathbf{p}) \\ E_3(\mathbf{p}) \end{pmatrix}$$

where  $\omega_{ij}(\mathbf{v}) = \nabla_{\mathbf{v}} E_i \cdot E_j(\mathbf{p})$ . Then each  $\omega_{ij}$  is a 1-form,  $\omega_{ij} = -\omega_{ji}$ , and the 1-forms are called **connected forms** of the frame field  $E_1, E_2, E_3$ . Notice that  $\omega_{ij} = 0$  for  $i = j$  and so the matrix will be skew-symmetric.

Also note that for any vector field  $V$  on  $\mathbb{R}^3$ ,

$$\nabla_V E_i = \sum_j \omega_{ij}(V) E_j \quad (1 \leq i \leq 3).$$

*How to Find Connection Forms?* Suppose  $E_1, E_2, E_3$  are arbitrary frame fields on  $\mathbb{R}^3$ . First we express these in terms of natural fields  $U_1, U_2, U_3$  as follows:

$$\begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} U_1 \\ U_2 \\ U_3 \end{pmatrix}$$

where each  $a_{ij} = E_i \cdot U_j \in \mathbb{R}$  and the matrix  $A$  is known as the **attitude matrix** of the frame field.

**Theorem 0.13.** Let  $A$  be the attitude matrix and  $\omega$  be the matrix of connection forms of a frame field  $E_1, E_2, E_3$ . Then

$$\omega = -A^\top dA$$

where  $dA = (da_{ij})$ . Equivalently,

$$\omega_{ij} = \sum_k a_{jk} da_{ik} \quad 1 \leq i, j \leq 3.$$

**Choice of Connection & Frame Bundle.** To understand a natural, intrinsic way to describe the geometry of a space curve  $\gamma : [0, 1] \rightarrow \mathbb{R}^3$  where  $\gamma(0) = \gamma(1)$  which in this case is our smooth manifold  $S^1$ . We can use the Frenet-Serret Frame. Where we define the unit tangent  $\hat{\mathbf{T}} = \gamma'$ , normal  $\hat{\mathbf{N}}$ , and binormal  $\hat{\mathbf{B}}$  vectors, satisfying the differential equation:

$$\frac{d}{ds} \begin{pmatrix} \hat{\mathbf{T}} \\ \hat{\mathbf{N}} \\ \hat{\mathbf{B}} \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{T}} \\ \hat{\mathbf{N}} \\ \hat{\mathbf{B}} \end{pmatrix}$$

Where  $\kappa = |\gamma''|$  and  $\tau = \frac{(\gamma''' \times \gamma') \cdot \gamma'}{|\gamma''|^2}$  are the curvature and torsion of our manifold.

However, there are other choices for an orthonormal basis, reason why we did choose this is because this frame does not exist if there exist points on the curve where  $\kappa = 0$ . The other choice is the Bishop Relatively Parallel Adapted Frame (RPAF). Suppose there exist an orthonormal basis for  $\gamma$  where we define the unit tangent vector  $\hat{\mathbf{T}}$ , a new vector called the relative parallel vector field  $\mathbf{M}_1$ . Defined such that it's a normal vector field (i.e.  $\mathbf{M}_1 \cdot \hat{\mathbf{T}} = 0$ ) and its derivative is parallel to  $\hat{\mathbf{T}}$  (i.e.  $\mathbf{M}_1' \cdot \mathbf{M}_2 = 0$ ).

Before that let's form an arbitrary unit orthonormal frame,  $\hat{\mathbf{T}}$ ,  $\hat{\mathbf{N}}_1$  and  $\hat{\mathbf{N}}_2$ , then we have:

$$\frac{d}{ds} \begin{pmatrix} \hat{\mathbf{T}} \\ \hat{\mathbf{N}}_1 \\ \hat{\mathbf{N}}_2 \end{pmatrix} = \begin{pmatrix} 0 & c_{12} & c_{13} \\ -c_{12} & 0 & c_{23} \\ -c_{13} & -c_{23} & 0 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{T}} \\ \hat{\mathbf{N}}_1 \\ \hat{\mathbf{N}}_2 \end{pmatrix}$$

Now, it's trivial that  $\mathbf{M}_1$  is a linear combination of  $\hat{\mathbf{N}}_1$  and  $\hat{\mathbf{N}}_2$ , so to find this normal vector field to be relatively parallel, there exist this smooth angle function  $\theta$  such that  $\mathbf{M}_1 = L(\cos \theta \hat{\mathbf{N}}_1 + \sin \theta \hat{\mathbf{N}}_2)$  where  $L$  just shows constant length. For our condition to hold,  $\mathbf{M}_1$  must be parallel to the tangent when derived. [Bis75]

$$\mathbf{M}_1' = L((\theta' + c_{23})(-\sin \theta \hat{\mathbf{N}}_1 + \cos \theta \hat{\mathbf{N}}_2) - (c_{12} \cos \theta + c_{13} \sin \theta)\hat{\mathbf{T}})$$

From this  $\mathbf{M}_1$  is relatively parallel if and only if  $\theta' = -c_{23}$  Using this information, the bishop frame is shown as:

$$\frac{d}{ds} \begin{pmatrix} \hat{\mathbf{T}} \\ \hat{\mathbf{M}}_1 \\ \hat{\mathbf{M}}_2 \end{pmatrix} = \begin{pmatrix} 0 & k_1 & k_2 \\ -k_1 & 0 & 0 \\ -k_2 & 0 & 0 \end{pmatrix} \begin{pmatrix} \hat{\mathbf{T}} \\ \hat{\mathbf{M}}_1 \\ \hat{\mathbf{M}}_2 \end{pmatrix}$$

From comparing  $\hat{\mathbf{T}}$  from Frenet-Serret Frame and and bishop frame,  $\kappa \hat{\mathbf{N}} = k_1 \hat{\mathbf{M}}_1 + k_2 \hat{\mathbf{M}}_2 \implies \kappa^2 = k_1^2 + k_2^2$ .  $\hat{\mathbf{N}}$  can be written as a linear combination of  $\hat{\mathbf{M}}_1$  and  $\hat{\mathbf{M}}_2$ . Using this we can show that:

$$\hat{\mathbf{N}} = \cos \theta \hat{\mathbf{M}}_1 + \sin \theta \hat{\mathbf{M}}_2$$

and differentiating shows that:

$$\hat{\mathbf{N}}' = -\kappa \hat{\mathbf{T}} + \tau \hat{\mathbf{B}} = -\kappa \hat{\mathbf{T}} + \theta'(-\sin \theta \hat{\mathbf{M}}_1 + \cos \theta \hat{\mathbf{M}}_2)$$

This shows that  $\theta' = \tau$ , hence showing that  $(\kappa, \int \tau ds)$  are the polar coordinates for the curve  $(k_1, k_2)$ . [Bis75]

## HOW TO APPROACH THE PROBLEM?

The smooth manifold  $S^1$  is embedded in the  $\mathbb{R}^3$ . Suppose we have a simple closed smooth curve parametrised by arc-length  $\gamma : [0, 1] \rightarrow S^1$  such that  $\gamma(0) = \gamma(1)$ . Given the base manifold  $S^1$ , define the vector fiber bundle  $(S^1, E, \pi)$  such that  $E = \mathbb{R}^3 \times S^1$ , and  $\pi : \mathbb{R}^3 \times S^1 \rightarrow S^1$ . We then require a choice of connection  $\nabla$  which is well defined where for  $\nabla_X Y$ ,  $X \in \Gamma(T_p S^1)$  and  $Y \in \Gamma(E)$ . This choice must be defined such that in a desirable orthonormal frame  $F = (F_1, F_2, F_3)$ ,  ${}_X F_i = 0^2$  for  $i = 1, 2, 3$ .

<sup>2</sup>In the case of  $S^1$ ,  $X$  is really just  $\hat{\mathbf{T}}(p)$  the tangent vector at any point  $p$ . I will continue to refer to this as  $X$ .

Take a natural choice of an orthonormal frame basis  $(E_1(s) \ E_2(s) \ E_3(s))^\top$ . The connection form matrix then defines

$$\begin{pmatrix} \nabla_X E_1 \\ \nabla_X E_2 \\ \nabla_X E_3 \end{pmatrix} = \begin{pmatrix} \omega_{11}(X) & \omega_{12}(X) & \omega_{13}(X) \\ \omega_{21}(X) & \omega_{22}(X) & \omega_{23}(X) \\ \omega_{31}(X) & \omega_{32}(X) & \omega_{33}(X) \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix}$$

or equivalently we write  $\nabla_X E = \omega E$ . Now in another choice of frame  $F = AE$ , where  $A$  is an attitude matrix,  $\nabla_X F = \tilde{\omega} F$ . Then

$$\begin{aligned} \nabla_X F &= \tilde{\omega} F = \tilde{\omega} A E = \nabla_X (A E) \\ &= (\nabla_X A) E + A (\nabla_X E) \\ &= dA E + A \omega E \\ &= [dA + A \omega] E \end{aligned}$$

The correct choice of connection in the frame  $F$  requires  $\nabla_X F = 0 \iff \tilde{\omega} = 0 \iff \omega = -A^{-1} dA = -A^\top dA$ . The last equality holds due to the orthonormality of  $F$  wherefore  $A^{-1} = A^\top$ .

Following the former,  $\omega = -A^\top dA = -A^\top (dA A^\top) A = -A^\top C A$  where the matrix  $C$  is known as the ‘Maurer-Cartan form’. And so as defined by the previous section,

$$\frac{dA}{ds} = -A \omega = A A^\top C A = C A.$$

So we can define  $C := \frac{dA}{ds} A^{-1}$ , or

$$C := \begin{pmatrix} \dots & F'_1 & \dots \\ \dots & F'_2 & \dots \\ \dots & F'_3 & \dots \end{pmatrix} \begin{pmatrix} \dots & F_1 & \dots \\ \dots & F_2 & \dots \\ \dots & F_3 & \dots \end{pmatrix}^{-1} = \begin{pmatrix} F'_1 \cdot F_1 & F'_1 \cdot F_2 & F'_1 \cdot F_3 \\ F'_2 \cdot F_1 & F'_2 \cdot F_2 & F'_2 \cdot F_3 \\ F'_3 \cdot F_1 & F'_3 \cdot F_2 & F'_3 \cdot F_3 \end{pmatrix}$$

Using this method, one can derive the Frenet-Serret or Bishop frames from the  $C$  matrix, or vice-versa.

## DISCUSSION

**Commentary on Bhattacharjee et. al.** Here we return to [BST92] to generalise the process of finding the rotational transform when give a choice of frame and connection. Suppose the base manifold is  $S^1$ , and a general frame field  $(\mathbf{T}(s) \ \mathbf{N}_2(s) \ \mathbf{N}_3(s))$  where all three vectors are orthonormal. Following Eq. (3) from [BST92], a point on the magnetic surface can be described in terms of the local coordinates as

$$\mathbf{x}(\rho, \theta, s) = \mathbf{X}(s) + \rho \cos \theta \ \mathbf{N}_2(s) + \rho \sin \theta \ \mathbf{N}_3(s) \quad (1)$$

where  $\mathbf{X} = \mathbf{X}(s)$  is the magnetic axis and  $\mathbf{X}'(s) = \mathbf{T}(s)$ . If we assume a generalised Maurer-Cartan form

$$C = \begin{pmatrix} 0 & k_1 & k_2 \\ -k_1 & 0 & k_3 \\ -k_2 & -k_3 & 0 \end{pmatrix}, \quad (2)$$

the covariant vectors can be written as

$$\frac{\partial \mathbf{x}}{\partial \rho} = \cos \theta \ \mathbf{N}_2(s) + \sin \theta \ \mathbf{N}_3(s) \quad (3)$$

$$\frac{\partial \mathbf{x}}{\partial \theta} = -\rho \sin \theta \ \mathbf{N}_2(s) + \rho \cos \theta \ \mathbf{N}_3(s) \quad (4)$$

$$\begin{aligned} \frac{\partial \mathbf{x}}{\partial s} &= \mathbf{T}(s) + \rho \cos \theta \ \mathbf{N}'_2(s) + \rho \sin \theta \ \mathbf{N}'_3(s) \\ &= \mathbf{T}(s) + \rho \cos \theta (-k_1 \mathbf{T}(s) + k_3 \mathbf{N}_3(s)) + \rho \sin \theta (-k_2 \mathbf{T}(s) - k_3 \mathbf{N}_2(s)) \\ &= (1 - k_1 \rho \cos \theta - k_2 \rho \sin \theta) \mathbf{T}(s) - k_3 \rho \sin \theta \mathbf{N}_2(s) + k_3 \rho \cos \theta \mathbf{N}_3(s). \end{aligned} \quad (5)$$

Hence the coefficients of the first fundamental form can be written as

$$g_{ij} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & \rho^2 k_3 \\ 0 & \rho^2 k_3 & (1 - \rho h(\theta, s))^2 + \rho^2 k_3^2 \end{pmatrix} \quad (6)$$

where  $h(\theta, s) = k_1 \cos \theta + k_2 \sin \theta$ . Consequentially after some rearrangement the metric can be written as

$$dx^2 = d\rho^2 + \rho^2(d\theta + k_3 ds)^2 + (1 - \rho h)^2 ds^2 \quad (7)$$

If we introduce the crucial parameter

$$\omega = \theta + \int k_3 ds, \quad (8)$$

the metric can be simplified to  $dx^2 = d\rho^2 + \rho^2 d\omega^2 + (1 - \rho h)^2 ds^2$ . Although we will not implement this yet, we will discuss this important substitution later.

In the system's configuration, the tangent to a magnetic field line, or a particle's trajectory in the magnetic confinement  $\mathbf{x}(s)$  is everywhere parallel to the magnetic field  $\mathbf{B} = (B^\rho \ B^\theta \ B^s)$ . i.e  $d\mathbf{x} \times \mathbf{B} = 0$ . More concretely, this means that

$$\frac{d\mathbf{x}}{ds} = \lambda(s)\mathbf{B}(\mathbf{x}(s)) \quad (9)$$

where  $\lambda(s)$  is an arbitrary scalar function. This means that

$$\frac{dx^i}{ds} = \lambda B^{x^i} \quad \text{for } x^i = \rho, \theta, s. \quad (10)$$

The upper indices of  $\mathbf{B}$  can be written in terms of lower indices as  $B^i = g^{ij} B_j$ , where

$$g^{ij} = \frac{1}{(1 - \rho h(\theta, s))^2} \begin{pmatrix} (1 - \rho h(\theta, s))^2 & 0 & 0 \\ 0 & \frac{(1 - \rho h(\theta, s))^2}{\rho^2} + k_3^2 & -k_3 \\ 0 & -k_3 & 1 \end{pmatrix}. \quad (11)$$

And so the upper indices are

$$B^\rho = B_\rho,$$

$$B^\theta = \left[ \frac{1}{\rho^2} + \frac{k_3^2}{(1 - \rho h)^2} \right] B_\theta - \frac{k_3}{(1 - \rho h)^2} B_s,$$

$$B^s = -\frac{k_3}{(1 - \rho h)^2} B_\theta + \frac{1}{(1 - \rho h)^2} B_s.$$

With this preparation, we can now write

$$\frac{d\theta}{ds} = \frac{B^\theta}{B^s} = \frac{(1 - \rho h(\theta, s))^2 B_\theta}{\rho^2 (B_s - k_3 B_\theta)} - k_3. \quad (12)$$

Using this, the rotational transform  $\iota$  can be written as

$$\iota = \oint d\theta = \oint \frac{d\theta}{ds} ds \quad (13)$$

$$= \oint \left[ \frac{(1 - \rho h(\theta, s))^2 B_\theta}{\rho^2 (B_s - k_3 B_\theta)} - k_3 \right] ds \quad (14)$$

$$= \oint \frac{(1 - \rho h(\theta, s))^2 B_\theta}{\rho^2 (B_s - k_3 B_\theta)} ds - \oint k_3 ds \quad (15)$$

$$\equiv \iota_m - \iota_g \quad (16)$$

Now one may ask, "what is the significance of this?". Equation (15) shows a generalisation of the rotational transform, and its splitting into dynamical and geometric phase as discussed in [BST92] in the case where  $k_3 = \tau$ . The choice of equation (8), as used in [BST92] leads to the geometric phase  $\iota_g = \int \tau ds$  by removing the  $k_3$  term in the dynamical phase of equation (15) and hence decoupling the constant from the other terms. It is worth mentioning that with an appropriate choice of connection, the metric tensor  $g_{ij}$  can become diagonal, thus resulting in  $k_3 = 0$ . This is precisely what happens in the case of the Bishop frame-adapted connection with  $C^{Bishop}$ . Then by choosing the Frenet-Serret frame, as done by [BST92], the current  $\iota_g$  result is rederived. It is worthy to question why  $\iota_g$  takes this particular form. To this then we may answer:

*"The geometric phase  $\iota_g$  takes its form when a choice of Bishop or Relatively Parallel frame-appropriate connection is applied on a Frenet-Serret frame. Otherwise, the geometric phase will be absorbed in to the dynamical phase  $\iota_m$ ".*

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